Power BI Public School Catchments Repository

This public repository stores school catchment reference data and processing outputs used by the Power BI applicant mapping report.

It is designed to provide public, reusable spatial reference layers for:

- Scotland secondary non-denominational catchments
- Scotland secondary denominational / Roman Catholic catchments
- School-centred fallback GeoJSON layers
- Power BI school reference lookup tables

The private applicant-processing repository uses this public repository as a read-only source of catchment polygons and lookup tables.

Private applicant-level data must not be uploaded to this repository.

Purpose

This repository supports the Power BI mapping report by hosting public GeoJSON files that can be loaded dynamically by Azure Maps reference layers.

The repository currently supports:

1. National Scotland secondary non-denominational catchment polygons.
2. National Scotland secondary denominational / Roman Catholic catchment polygons.
3. Combined per-school denominational catchment GeoJSON files for multi-part or cross-authority catchments.
4. Fallback GeoJSON files for schools without a real catchment polygon.
5. A Power BI-ready lookup table mapping each school to its catchment or fallback GeoJSON URL.
6. Match-review and processing-summary outputs for QA.

Data privacy

This repository is public.

It should contain only public reference data, such as:

- school catchment GeoJSON files
- fallback GeoJSON files
- school reference lookup CSV files
- match-review files
- processing summaries
- public school location/reference files

Do not upload applicant-level records, applicant postcodes, applicant coordinates, or private recruitment data to this repository.

Applicant data belongs in the private applicant-processing repository.

Repository structure

```
powerbi-school-catchments/
|
|-- incoming/
| |-- scotland-secondary-non-denom/
| | |-- non_denom_pub_schlsn.shp
| | |-- non_denom_pub_schlsn.shx
| | |-- non_denom_pub_schlsn.dbf
| | |-- non_denom_pub_schlsn.prj
| |
| |-- scotland-secondary-denom/
| |-- denom_pub_schlsd.shp
| |-- denom_pub_schlsd.shx
| |-- denom_pub_schlsd.dbf
| |-- denom_pub_schlsd.prj
|
|-- inputs/
| |-- scottish_secondary_schools_for_mapping.csv
| |-- school_reference_layer_lookup_with_fallback_and_fife_catchments.csv
| |-- scotland_sn_manual_catchment_overrides.csv
| |-- scotland_sd_manual_catchment_overrides.csv
|
|-- scripts/
| |-- inspect_shapefile_schema.py
| |-- process_scotland_sn_catchments.py
| |-- process_scotland_sd_catchments.py
|
|-- scotland-secondary-non-denom-catchments/
| |-- one GeoJSON file per source non-denominational catchment
|
|-- scotland-secondary-denom-catchments/
| |-- one GeoJSON file per source denominational catchment
| |-- combined-by-school/
| |-- one combined GeoJSON file per matched denominational school
|
|-- school-fallbacks/
| |-- school-centred fallback GeoJSON files
```

|

|-- outputs/

| |-- scotland-secondary-non-denom/

| |-- scotland-secondary-denom/

| |-- other processing outputs

|

|-- .github/

|-- workflows/

|-- inspect_scotland_sn_catchments.yml

|-- inspect_scotland_sd_catchments.yml

|-- process_scotland_sn_catchments.yml

|-- process_scotland_sd_catchments.yml

Main inputs

1. School mapping input

The main school mapping input is:

inputs/scottish_secondary_schools_for_mapping.csv

This file should contain the school-level location and reference data required for matching catchments and building school map points.

Important columns include:

- SchoolName
- PostCode_clean
- LAName
- SeedCode
- Latitude
- Longitude
- Postcode_area
- Postcode_district

2. School reference layer lookup input

The catchment processing stages build on an existing lookup table, for example:

inputs/school_reference_layer_lookup_with_fallback_and_fife_catchments.csv

This table links school keys to fallback GeoJSON URLs and is progressively enriched with real catchment GeoJSON URLs.

Important columns include:

- SchoolKey
- CatchmentGeoJsonUrl
- FallbackGeoJsonUrl
- Postcode_area
- Postcode_district
- HasRealCatchmentGeoJson

3. Secondary non-denominational shapefile

The source files are stored in:

incoming/scotland-secondary-non-denom/

The expected basename is:

non_denom_pub_schlsn

The required shapefile components are normally:

- non_denom_pub_schlsn.shp
- non_denom_pub_schlsn.shx
- non_denom_pub_schlsn.dbf
- non_denom_pub_schlsn.prj

Optional accompanying files may include:

- .cpg
- .cst
- .xml

4. Secondary denominational shapefile

The source files are stored in:

incoming/scotland-secondary-denom/

The expected basename is:

denom_pub_schlsd

The required shapefile components are normally:

- denom_pub_schlsd.shp
- denom_pub_schlsd.shx
- denom_pub_schlsd.dbf
- denom_pub_schlsd.prj

Optional accompanying files may include:

- .cpg
- .cst
- .xml

Main workflows

1. Inspect Scotland secondary non-denominational catchments

Workflow:

.github/workflows/inspect_scotland_sn_catchments.yml

Purpose:

- inspect the non-denominational shapefile schema
- confirm record count and fields
- produce schema summary outputs

Typical outputs:

outputs/scotland-secondary-non-denom-schema/schema_report.md
outputs/scotland-secondary-non-denom-schema/schema_fields.csv
outputs/scotland-secondary-non-denom-schema/sample_records.csv
outputs/scotland-secondary-non-denom-schema/candidate_field_distinct_summary.csv

2. Process Scotland secondary non-denominational catchments

Workflow:

.github/workflows/process_scotland_sn_catchments.yml

Script:

scripts/process_scotland_sn_catchments.py

Purpose:

- read the national non-denominational secondary shapefile
- transform geometry from British National Grid to WGS84
- split the shapefile into one GeoJSON file per source catchment record
- match catchments to schools using seed code, school name, local authority, and manual overrides
- update the school reference lookup with real catchment URLs
- write match-review and processing-summary outputs

Important outputs:

outputs/scotland-secondary-non-denom/school_reference_layer_lookup_plus_scotland_sn.csv outputs/scotland-secondary-non-denom/scotland_sn_catchment_match_review.csv

outputs/scotland-secondary-non-denom/scotland_sn_catchment_geojson_index.csv

outputs/scotland-secondary-non-denom/processing_summary.md

scotland-secondary-non-denom-catchments/*.geojson

3. Inspect Scotland secondary denominational catchments

Workflow:

.github/workflows/inspect_scotland_sd_catchments.yml

Purpose:

- inspect the denominational shapefile schema
- confirm record count and fields
- produce schema summary outputs

Typical outputs:

outputs/scotland-secondary-denom-schema/schema_report.md

outputs/scotland-secondary-denom-schema/schema_fields.csv

outputs/scotland-secondary-denom-schema/sample_records.csv

outputs/scotland-secondary-denom-schema/candidate_field_distinct_summary.csv

4. Process Scotland secondary denominational catchments

Workflow:

.github/workflows/process_scotland_sd_catchments.yml

Script:

scripts/process_scotland_sd_catchments.py

Purpose:

- read the national denominational secondary shapefile
- transform geometry from British National Grid to WGS84
- split the shapefile into one GeoJSON file per source catchment record
- match catchments to schools using seed code, school name, local authority, and manual overrides
- generate combined per-school denominational GeoJSON files
- update the school reference lookup with combined per-school denominational catchment URLs
- write match-review and processing-summary outputs

Important outputs:

outputs/scotland-secondary-denom/school_reference_layer_lookup_plus_scotland_sn_sd.csv

outputs/scotland-secondary-denom/scotland_sd_catchment_match_review.csv outputs/scotland-secondary-denom/scotland_sd_catchment_geojson_index.csv

outputs/scotland-secondary-denom/scotland_sd_combined_by_school_index.csv

outputs/scotland-secondary-denom/processing_summary.md

scotland-secondary-denom-catchments/*.geojson

scotland-secondary-denom-catchments/combined-by-school/*.geojson

The final lookup for Power BI is:

outputs/scotland-secondary-denom/school_reference_layer_lookup_plus_scotland_sn_sd.csv

Manual override files

Some source catchment names do not exactly match school names in the school mapping file.

Manual overrides are stored in:

inputs/scotland_sn_manual_catchment_overrides.csv

inputs/scotland_sd_manual_catchment_overrides.csv

These files map a source catchment row to a target school name and local authority.

Example columns:

- source_row
- source_school_nam
- source_local_auth
- target_school_name
- target_local_auth
- override_reason

Manual overrides should be conservative and auditable.

They are useful for:

- shortened school names
- apostrophe or punctuation differences
- RC wording variants
- descriptive catchment labels
- cross-authority denominational catchment labels

Manual overrides should not be used to force genuinely ambiguous multi-school or choice-area records into one school unless the assignment is known and defensible.

Current successful processing results

Scotland secondary non-denominational processing

The non-denominational processing stage produced:

- Catchment records processed: 324
- Matched catchments after manual overrides: 305
- Unmatched catchments after manual overrides: 19
- Seed-code matches: 278
- Manual override matches: 23
- Manual override target not found: 0

The SN output significantly increased the number of schools with real catchment polygons.

Scotland secondary denominational processing

The denominational processing stage produced:

- Catchment records processed: 63
- Row-level GeoJSON files created: 63
- Combined per-school GeoJSON files created: 54
- Schools with multiple denominational catchment features: 8
- Matched catchments: 63
- Unmatched catchments: 0
- Seed-code matches: 53
- Manual override matches: 9
- Manual override target not found: 0
- Manual override multiple matches: 0
- School name + local authority matches: 1
- School name only matches: 0
- Lookup rows with real catchment URL after update: 350

The final denominational workflow creates combined per-school GeoJSON files to avoid losing cross-authority or multi-part denominational catchments.

Final Power BI lookup

The final lookup produced by this repository is:

outputs/scotland-secondary-denom/school_reference_layer_lookup_plus_scotland_sn_sd.csv

When copied into the private applicant-processing workflow, this file is renamed for Power BI as:

outputs/powerbi_school_reference_layer_lookup.csv

This table allows Power BI to select the correct GeoJSON URL dynamically for the selected school.

How the private applicant pipeline uses this repository

The private applicant-processing workflow checks out this public repository during GitHub Actions.

It uses:

- scotland-secondary-non-denom-catchments/*.geojson
- scotland-secondary-denom-catchments/combined-by-school/*.geojson
- outputs/scotland-secondary-denom/school_reference_layer_lookup_plus_scotland_sn_sd.csv

The private workflow then:

1. standardises applicant spreadsheets
2. matches Scottish postcodes to ONSPD
3. assigns applicants to SN and SD catchments using these public GeoJSON files
4. builds Power BI-ready applicant and school map outputs

Applicant data is never written to this public repository.

Updating the public catchment repository

Use this process when new school catchment shapefiles are available.

Step 1 - Upload updated shapefile components

For non-denominational catchments, update:

incoming/scotland-secondary-non-denom/

For denominational catchments, update:

incoming/scotland-secondary-denom/

Upload all required shapefile components, especially:

- .shp
- .shx
- .dbf
- .prj

Step 2 - Run schema inspection

Run the relevant inspection workflow:

- Inspect Scotland Secondary Non-Denominational Catchments
- Inspect Scotland Secondary Denominational Catchments

Review:

- schema_report.md
- candidate_field_distinct_summary.csv

Confirm the fields still include:

- local_auth
- school_nam
- seed_code
- la_s_code

Step 3 - Run processing workflow

Run the relevant processing workflow:

- Process Scotland Secondary Non-Denominational Catchments
- Process Scotland Secondary Denominational Catchments

Step 4 - Review summary and match-review files

Review:

- processing_summary.md
- catchment_match_review.csv
- catchment_geojson_index.csv

Look for:

- matched catchment count
- unmatched catchment count
- manual override target not found
- duplicate or multi-school cases

Step 5 - Update manual overrides if required

If obvious name mismatches appear, update the relevant override file:

inputs/scotland_sn_manual_catchment_overrides.csv

inputs/scotland_sd_manual_catchment_overrides.csv

Then rerun the workflow.

Step 6 - Confirm final lookup

The final lookup should be:

outputs/scotland-secondary-denom/school_reference_layer_lookup_plus_scotland_sn_sd.csv

This is the lookup consumed by the private applicant-processing repo and by Power BI.

QA checklist after processing

After processing catchments, confirm:

- GeoJSON files were created
- processing_summary.md updated
- match-review CSV updated
- expected number of catchments were processed
- manual override target not found count is 0, or explainable
- final lookup contains real catchment URLs
- denominational combined-by-school outputs were created
- public raw GitHub URLs open successfully

Recommended spot checks in Power BI:

Non-denominational schools:

- Inverkeithing High School
- Boroughmuir High School
- Currie Community High School
- Queensferry Community High School
- The Royal High School
- Hyndland Secondary School

Denominational / RC schools:

- Holy Rood RC High School
- St Augustine's High School
- St Thomas Of Aquin's High School
- St Kentigern's Academy
- St Modan's High School
- St Andrew's R C High School
- Holyrood Secondary School
- Notre Dame High School

For St Modan's High School, confirm the combined multi-part denominational catchment appears.

Important design notes

Coordinate reference system

The source Scotland shapefiles are assumed to be in British National Grid:

EPSG:27700

The processing scripts transform the geometries to WGS84:

EPSG:4326

This is required for Azure Maps and web GeoJSON use.

Row-level versus combined GeoJSON files

The non-denominational workflow currently produces row-level catchment GeoJSON files.

The denominational workflow produces both:

- row-level GeoJSON files
- combined per-school GeoJSON files

The combined per-school denominational files are important because denominational catchments can span multiple local authority areas or appear in multiple source records.

Power BI should use the combined denominational file where available.

SchoolKey format

The standard school key format is:

SchoolName|Postcode

Example:

Holy Rood RC High School|EH15 3ST

This key is used to connect:

- school selector rows
- school reference lookup rows
- matched catchment records
- applicant catchment membership fields

Common troubleshooting

Workflow succeeds but outputs do not update

Check that the script being run is the updated script.

Search the script for expected text such as:

- manual_override_matches
- combined-by-school
- school_reference_layer_lookup_plus_scotland_sn_sd.csv

Also confirm the workflow ran on the main branch.

Manual overrides do not apply

Check:

- override CSV path in the workflow
- source_row values match current match-review row numbers
- target_school_name exactly matches the school mapping file after normalisation
- target_local_auth matches the school mapping local authority

Review:

- manual override matches
- manual override target not found
- manual override multiple matches

in processing_summary.md.

A school shows only part of its denominational catchment

Check that Power BI is using the final SN+SD lookup:

outputs/scotland-secondary-denom/school_reference_layer_lookup_plus_scotland_sn_sd.csv

Also check that the denominational workflow created:

scotland-secondary-denom-catchments/combined-by-school/

and that the selected school points to a combined-by-school URL where appropriate.

Catchment polygons fail to render in Azure Maps

Check that the URL is a raw GitHub URL, for example:

https://raw.githubusercontent.com/McCrackenLab/powerbi-school-catchments/refs/heads/main/...

Do not use the normal GitHub webpage URL.

Also check that the GeoJSON opens directly in a browser and that the file contains valid FeatureCollection JSON.

The private applicant pipeline fails to find public catchment files

Check that this repository still contains:

scotland-secondary-non-denom-catchments/
scotland-secondary-denom-catchments/combined-by-school/
outputs/scotland-secondary-denom/school_reference_layer_lookup_plus_scotland_sn_sd.csv

If folder names change, update the private applicant-processing scripts.

Recommended maintenance process

When new catchment data is released:

1. Upload new shapefile files to incoming/.
2. Run schema inspection workflow.
3. Review schema outputs.
4. Run processing workflow.
5. Review processing summary and match-review CSV.
6. Update manual overrides if required.
7. Rerun processing workflow.
8. Confirm final lookup output.
9. Refresh or rerun private applicant-processing workflow.
10. Refresh Power BI.

Files used by Power BI and private processing

The most important public output is:

outputs/scotland-secondary-denom/school_reference_layer_lookup_plus_scotland_sn_sd.csv

The most important public GeoJSON folders are:

scotland-secondary-non-denom-catchments/
scotland-secondary-denom-catchments/combined-by-school/

These should be kept stable where possible.

Notes for future development

Potential future improvements include:

- combined per-school GeoJSON files for non-denominational catchments as well as denominational catchments
- automated QA comparing yearly catchment counts
- stronger duplicate-school handling
- versioned releases for each catchment-processing refresh
- archive folders for old catchment outputs

Maintenance checklist

Before relying on a refreshed public catchment dataset, confirm:

- inspection workflow has run successfully
- processing workflow has run successfully
- processing summaries look sensible
- final lookup exists
- manual override failures are zero or explainable
- raw GitHub GeoJSON URLs are accessible
- Power BI can load the selected catchment polygon
- private applicant-processing workflow can still assign applicants to catchments